

Lesson 1-1: Combustion and Air Worksheet

Model Answers



A. Choose the correct answer.

1. Which condition allows a candle to keep burning?
   a) No air supply b) **Continuous fresh air**
   c) A closed lid d) Less oxygen
   b

2. In the experiment, which setup provides the least fresh air to the candle?
   a) Jar without a lid b) Jar with a small gap
   c) **Completely closed jar** d) Open candle
   c

3. Making a small gap at the bottom of the jar causes _____.
   a) **some fresh air to enter** b) all the air to leave
   c) the candle to go out immediately
   d) the oxygen to disappear
   a

4. Which flame would you expect to be the strongest?
   a) Flame in a completely closed jar
   b) Flame in a jar with a small gap
   c) **Flame in a completely open jar**
   d) Flame with no oxygen
   c

5. Which of the following best describes the relationship between air supply and combustion?
   a) **More fresh air supports stronger combustion.**
   b) Less fresh air makes combustion stronger.
   c) Air has no effect on combustion.
   d) Combustion removes the need for oxygen.
   a



B. Write True or False.

1. A candle can keep burning without oxygen. (**False**)
2. A small opening allows some fresh air to reach the candle. (**True**)
3. The amount of air available can affect the strength of a candle flame. (**True**)
4. The flame is strongest when the jar is completely sealed. (**False**)
5. During combustion, oxygen in the surrounding air is consumed. **TRUE**



C. Complete the following.

1. The process in which substances burn is called **combustion**.
2. During combustion, **oxygen** in the air is gradually consumed.
3. When there is not enough fresh air, combustion **stops**.
4. With only a small amount of fresh air, the candle burns with a **weaker** flame.
5. When the lid is removed completely, the flame becomes **stronger** and **brighter**.





D. What happens when...?



1. A candle is lit and left where fresh air can reach it freely?
   The candle keeps burning with a steady flame.

2. A burning candle is covered by a jar and the jar is completely closed?
   The flame goes out after a while.



3. The oxygen trapped inside a closed jar is gradually used up?
   The flame becomes weaker and eventually goes out.



4. A small opening is made at the bottom of the jar while the top remains covered?
   The candle continues burning with a weaker flame.



5. The lid is removed completely after the candle has been burning with only a small air gap?
   The flame becomes stronger and brighter.



E. Give reasons for the following



1. Oxygen is necessary for combustion.
   Because oxygen supports burning.

2. Fresh air must continuously flow in and out
   f
   Because fresh air supplies oxygen.

3. The flame in the completely closed jar goes out after a while.
   Because the oxygen inside the jar is used up, so combustion cannot continue.

4. The candle can still burn when there is a small gap at the bottom of the jar.
   Because fresh air can enter through the gap and supply oxygen.

5. The flame becomes brighter and stronger when the lid is completely removed.
   Because more fresh air and oxygen can reach the candle.







Lesson 1-2
Role of Oxygen



Try

1 (1) In the experiment of burning a standing candle inside a jar, what happens to the flame if you put a lid on the jar to stop air from coming in and out?

The flame goes out after a while.

(2) Why did the flame go out inside the jar with the lid? Choose the correct answer from below.



A: Because there was no water vapor in the jar.

B: Because it got dark inside the jar.

C: Because fresh air stopped coming in.

2 (1) What gas is included in the air and has the power to help things burn?

Oxygen.



(2) What happens to the burning when you put a lit incense stick into a jar of oxygen? Choose the correct answer from below.

A: It becomes weaker.

B: It burns brightly.

C: The flame goes out.



(3) What happens to the burning when you put a lit incense stick into a jar of carbon dioxide? Choose the correct answer from below.

A: It keeps burning as it is.

B: It burns brightly.

C: The flame goes out.



Exercise

1 (1) In the experiment of burning a candle inside a jar, what happens to the flame if you leave gaps at the top and bottom of the jar? Choose the correct answer from below.



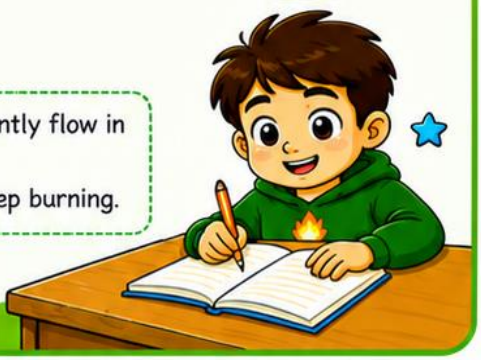
A: It keeps burning.

B: It goes out immediately.

(2) Fill in the blanks with the correct words.

For things to keep burning, fresh (1 oxygen) must constantly flow in and out.

In a place where air is not replaced, things (2 stop) keep burning.











Lesson 1-2: The Role of Oxygen



A. Choose the correct answer

1. Air is best described as:
   a) one pure gas
   b) a mixture of gases
   c) only oxygen
   d) only nitrogen

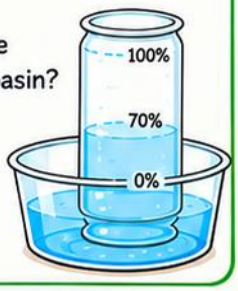
2. Which gas represents about one-fifth of the volume of air?
   a) Nitrogen
   b) Oxygen
   c) Carbon dioxide
   d) Water vapor

3. Which pair of gases does **NOT** support combustion?
   a) Oxygen and nitrogen
   b) Nitrogen and carbon dioxide
   c) Oxygen and carbon dioxide
   d) Oxygen and water vapor

4. Which tool carries gas from a gas cylinder to the collecting jar?
   a) Water basin
   b) Candle holder
   c) Bent tube
   d) Lid



5. About how much of the jar should be filled with gas before it is removed from the water basin?
   a) 10%
   b) 21%
   c) 50%
   d) 70%



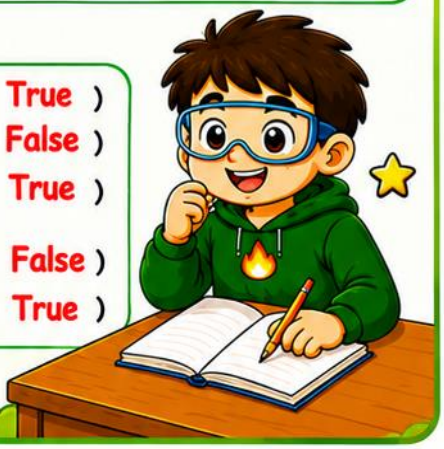
B. Complete the following

1. Air surrounds the **Earth** and is made up of several different gases.
2. Oxygen supports **combustion**, while nitrogen and carbon dioxide do not.
3. To collect a gas by water displacement, the gas jar is first completely filled with **water**.
4. As gas enters the inverted jar, it **pushes out** the water inside the jar.
5. During water displacement, the gas jar is placed **upside down** in the water basin.



C. Write True or False

1. Water vapor is one of the gases that may be found in air. (**True**)
2. Nitrogen is present in a smaller amount than oxygen in air. (**False**)
3. Equal amounts of gases should be used when comparing their effects on a candle. (**True**)
4. A collecting jar is placed upright in the basin during water displacement. (**False**)
5. The water level inside the collecting jar falls as gas enters it. (**True**)







Lesson 1-2: The Role of Oxygen



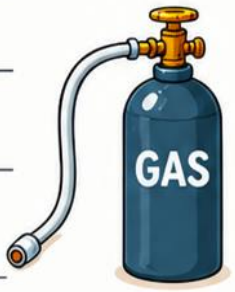
D. Write the scientific term

1	The process in which a substance burns.	Combustion
2	The gas that makes up the greatest percentage of air.	Nitrogen
3	The gas needed for combustion and for breathing:	Oxygen
4	The container used to collect a gas during the experiment.	Gas jar
5	The method in which a gas is collected by pushing water out of an inverted container.	Water displacement



E. What happens when...?

1	A lighted candle is placed in a jar containing oxygen?	The candle burns more brightly and strongly.
2	A lighted candle is placed in a jar containing nitrogen?	The flame goes out quickly.
3	A lighted candle is placed in a jar containing carbon dioxide?	The flame goes out.
4	Gas begins to enter a completely water-filled inverted jar through the bent tube?	The water is pushed out and the gas collects in the jar.
5	A burning substance is supplied with more oxygen than before?	It burns more brightly and strongly.



F. Give reasons for the following

1	Air is considered a mixture rather than a single substance.	Because it contains several different gases (nitrogen, oxygen, carbon dioxide and water vapor).
2	Equal amounts of oxygen, nitrogen, and carbon dioxide are prepared when comparing their effects on burning.	To make a fair comparison.
3	The gas jar must be inverted when gas is collected by water displacement.	To prevent the gas from escaping and to allow it to collect in the jar.
4	A lighted candle is useful for identifying which of the three gases supports combustion.	Because it burns in oxygen but goes out in nitrogen and carbon dioxide.
5	Oxygen is described as essential to living organisms as well as important in combustion.	Because it is needed for breathing in living organisms and also supports combustion.











Lesson 1-3: Air After Burning

Try

(1) Answer the following questions about the air after things burn.

① What gas has increased? **Carbon dioxide**

② What gas has decreased? **Oxygen**

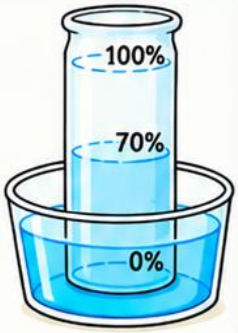


(2) What is the colorless and clear liquid used to check if there is carbon dioxide?

Limewater

Exercise

1 What happens when you put limewater in a jar with increased carbon dioxide and shake it? Choose the correct answer from below.



A: It turns blue.

B: It turns yellow.

C: It turns milky white.

2 How does the amount of oxygen change after burning?

It decreases.

3 How does the amount of carbon dioxide change after burning?

It increases.





Lesson 1-3: Air After Burning



A. Choose the Correct Answer

1 Which gas increases in the air after burning?

a) Oxygen **b) Carbon dioxide** c) Nitrogen d) Hydrogen



2 Which substance is used to test for carbon dioxide gas?

a) Distilled water **b) Limewater** c) Salt water d) Oil

3 What happens to limewater when carbon dioxide passes through it?

a) It turns blue. b) It turns green. **c) It turns milky white.** d) It remains colorless.

4 Which tool contains chemicals that react with certain gases and change color?

a) Gas detection tube b) Beaker c) Thermometer d) Combustion spoon

5 What is the main function of a sampling pump?

a) To heat a gas **b) To draw a gas sample through a detection tube**

c) To produce oxygen d) To measure temperature

6 Which device displays the percentage of a gas electronically?

a) Gas monitoring equipment b) Combustion spoon d) Safety goggles

7 Ancient Egyptians used bellows mainly to:

a) remove oxygen from a furnace. **b) push more air into a furnace.**

c) decrease the temperature of a fire. d) produce carbon dioxide.

8 Which of the following is a renewable source of energy?

a) Coal b) Natural gas **c) Sunlight** d) Petroleum

9 Increasing carbon dioxide in the atmosphere contributes to:

a) the greenhouse effect. b) less combustion.

c) lower oxygen use. d) the production of limewater.

10 A solar power plant helps protect the environment because it:

a) burns more fuel. **b) produces electricity without burning fossil fuels.**

c) increases carbon dioxide production. d) uses oxygen to produce electricity.







Lesson 1-3: Air After Burning



B. Write True or False

1. The amount of oxygen in air increases after burning. (False)
2. Limewater can be used as an indicator for the presence of carbon dioxide. (True)
3. Limewater remains completely clear when carbon dioxide is passed through it. (False)
4. Different gas-detection tubes may be designed to detect different gases. (True)
5. A gas-monitoring device can show gas concentration as a numerical percentage. (True)
6. Gas-monitoring equipment must always be broken after one use. (False)
7. Ancient Egyptians used combustion in some of their daily activities. (True)
8. Increasing the oxygen supply can make combustion occur faster. (True)
9. Renewable energy depends only on fossil fuels. (False)
10. Solar power can reduce some of the negative effects associated with carbon dioxide emissions. (True)



C. Write the Scientific Term

1. A process in which substances react with oxygen and burn. Combustion
2. The gas that is produced during burning and causes limewater to turn milky white. Carbon dioxide
3. A colorless, clear liquid used to test for the presence of carbon dioxide gas. Limewater
4. Small glass tubes containing chemicals that react with particular gases. Gas-detection tubes
5. A device used to draw a sample of gas through a gas-detection tube. Sampling pump
6. An electronic device with a sensor that measures the percentage of gas in the surrounding air. Gas-monitoring equipment
7. A blowing tool used to force additional air into a fire or furnace. Bellows
8. The rise in Earth's temperature caused by increasing carbon dioxide in the atmosphere. Greenhouse effect
9. Energy obtained from natural sources that do not run out. Renewable energy
10. A power plant that uses sunlight to generate electricity. Solar power plant













Lesson 1-3: Air After Burning



E. What Happens When...?



1. A burning candle is sealed inside a jar and left for some time?
   The flame goes out when the oxygen is used up.
2. A burning material receives a larger supply of fresh air? The oxygen available to a burning substance decreases?
   The burning decreases because less oxygen is available.
3. More oxygen is supplied to a burning fire?
   The fire burns faster and more brightly.
4. A gas sample reacts with the chemical inside a suitable gas-detection tube?
   The chemical changes color, indicating the presence of the gas.
5. A sampling pump is operated while connected correctly to a gas-detection tube?
   The pump draws a gas sample through the tube.
6. Bellows are used to push more air into a hot furnace?
   More air enters the furnace, making the fire burn more strongly.
7. The percentage of carbon dioxide in the atmosphere increases greatly?
   The greenhouse effect becomes stronger and Earth's temperature may rise.
8. Sunlight falls on the solar panels of a solar power plant?
   The solar panels convert sunlight into electrical energy.
9. Electricity is generated using solar energy instead of burning fossil fuels?
   Electricity is generated without burning fossil fuels.





Give Reasons for the Following

1. Air after combustion is different from air before combustion.
   Combustion consumes oxygen and produces carbon dioxide and water vapor.
2. Carbon dioxide increases in the air after some substances burn.
   Burning produces carbon dioxide.
3. Limewater is useful when investigating the gas produced during burning.
   Limewater turns milky when carbon dioxide passes through it.
4. Two jars—one with a burning candle and one without—can be used in the limewater experiment.
   The jar with the candle produces carbon dioxide, while the jar without a candle is the comparison.
5. Gas-detection tubes are useful for identifying gases in air.
   They contain chemicals that react with specific gases and change color.
6. A color chart is supplied with gas-detection tubes.
   It helps identify gases by matching the tube's color change with the chart.
7. Gas-monitoring equipment is useful in specialized hospitals and intensive care units.
   They measure and monitor oxygen and other gases in patients' breathing.
8. Ancient Egyptians used bellows when working with furnaces.
   Bellows force extra air into the furnace to help combustion.
9. Increasing atmospheric carbon dioxide can raise Earth's temperature.
   Carbon dioxide traps heat in the atmosphere, increasing Earth's temperature.
10. Using solar power can help reduce the negative effects of carbon dioxide.
    Solar power produces electricity without burning fossil fuels, reducing carbon dioxide emissions.











